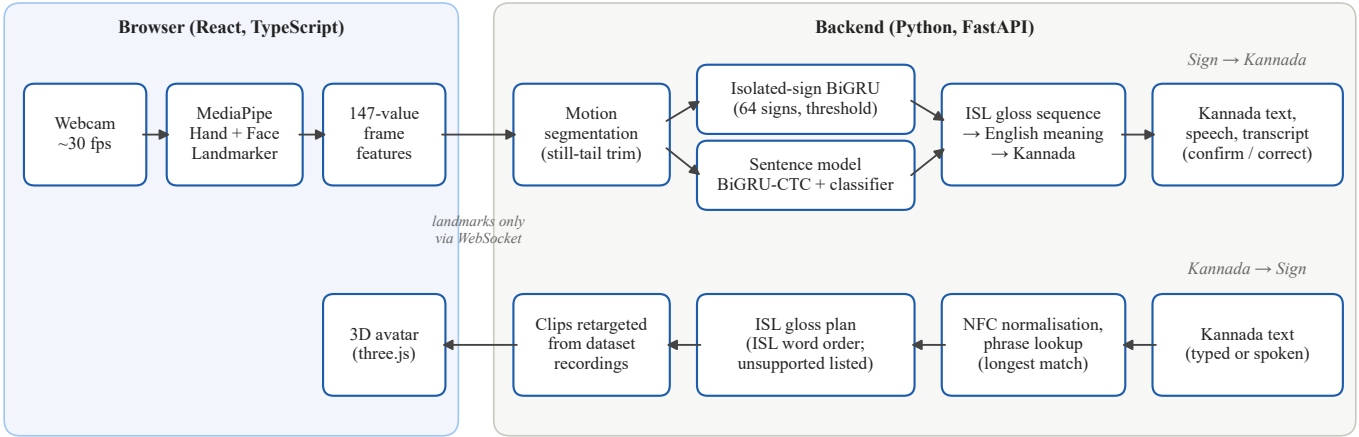

Browser (React, TypeScript)
Backend (Python, FastAPI)
Sign → Kannada
Webcam ~30 fps
MediaPipe Hand + Face Landmarker
147-value frame features
Motion segmentation (still-tail trim)
Isolated-sign BiGRU (64 signs, threshold)
Sentence model BiGRU-CTC + classifier
ISL gloss sequence → English meaning → Kannada
Kannada text, speech, transcript (confirm / correct)
landmarks only via WebSocket
Kannada → Sign
3D avatar (three.js)
Clips retargeted from dataset recordings
ISL gloss plan (ISL word order; unsupported listed)
NFC normalisation, phrase lookup (longest match)
Kannada text (typed or spoken)